Question: How does @Topology_v5.lean relates to the original essay?

The Lean code is a **partial formalization** of the essay — it implements Sections 3–4 faithfully but omits Sections 2 and 5 entirely. Here's the detailed mapping:

What's Implemented (Sections 3–4)

Essay Concept	Lean Code	Notes
Board graph G = (V, E)	‘boardGraph n : SimpleGraph (Fin n × Fin n)‘ with ‘adj‘	□ Exact match
Chain groups C^0, C^1 over $\mathbb{Z}_2$	‘C0 n := (Fin n × Fin n) → ZMod 2‘, ‘C1 n := edgeSet → ZMod 2‘	□ Exact match
Coboundary δ_0	‘delta_0 f e = f u + f v‘ via ‘edgeSum‘ and ‘Sym2.lift‘	□ Correctly handles undirected edges
Stone group indicator / capture condition (Prop 3.1)	‘isCaptured s c‘ + ‘prop_3_1‘	□ Same coboundary characterization; Lean version adds explicit occupancy hypothesis ‘hc‘ for technical clarity
Candidate eye definition	‘isCandidateEye s c e := s e = 0 ∧ ∀ u, Adj e u → u ∈ c‘ (in §4)	□ Matches essay’s ”orthogonal neighbors all in c”
Enclosure (flood-fill condition)	‘isEnclosed s R X := (∀ v ∈ R, s v = 0) ∧ (∀ boundary point → in X)‘	□ Exactly the flood-fill / coboundary condition described
Benson’s monotone operator f	‘benson_f s X‘ with monotonicity lemma	□ Same definition: stones whose liberties lie in regions enclosed by X
Knaster–Tarski greatest fixed point = unconditionally alive	‘unconditionally_alive s := OrderHom.gfp (benson_op s)‘	□ Direct implementation of Theorem 4.1
Two-eyes theorem (Lemma 4.2)	Full proof with case split on ‘h_libs‘	□ Complete; Lean version makes explicit the hypothesis that e_1, e_2 are the *only* liberties

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What's Omitted

Essay Section	Content	Status in Lean
Section 2 (State Space)	Legal positions P_n, move relation R, positional superko, game states (p, h), DAG/-poset structure, Proposition 2.1 & Corollary 2.2	□ Not present at all
Section 5 (CGT / Endgames)	Direct-sum decomposition of endgames, surreal numbers, infinitesimals, independence hypothesis	□ Not present at all

Section 0 acknowledgment details, Remark 3.2 on false eyes, Remark 4.3 on converse failure, Open problems	Conceptual discussion	□ Not formalized (expected)

Key Differences / Refinements in the Lean Version

1. **More explicit hypotheses:** Lemma 4.2's `h_libs` condition ("all liberties of c are e_1 or e_2") is stated explicitly, whereas the essay phrases it more informally as "remain enclosed throughout the iteration." This strengthens rigor.
2. **Restricted to Black stones:** The Lean `benson_f` and `unconditionally_alive` only consider Black stones (`s x = 1`). The essay's lattice L is described as "Black stones on the board," so this matches, but the Lean code doesn't generalize to arbitrary color.
3. **Prop 3.1 structure:** The Lean version adds (hc : ∀ v ∈ c, s v ≠ 0) (the group must actually be occupied) and reformulates the capture condition in terms of `isCaptured` + a universal quantification over edges — functionally equivalent but technically more careful.
4. **Proof completeness:** Lemma 4.2 is fully proved in Lean (including the $\{e_1\}$ / $\{e_2\}$ enclosure constructions), whereas the essay's proof is sketched informally.

Summary

The Lean code is a **faithful, rigorous implementation of the core algebraic-topological-heart** of the essay (Sections 3–4): coboundary-based liberties, Benson's operator, Knaster–Tarski life/death, and the two-eyes lemma. It deliberately leaves out:

- The game-theoretic superko/DAG framework (Section 2)
- The combinatorial game theory / surreal number endgame analysis (Section 5)

So: **the Lean code formalizes the "algebraic topology + lattice theory" pillar of the essay, but not its full scope.**